

Diet and Microbiome-Directed Therapy 2.0 for IBD

Ashwin N. Ananthakrishnan,^{1,2} Kevin Whelan,⁴ Jessica R. Allegretti,^{2,3} and Harry Sokol^{5,6,7}



¹Division of Gastroenterology, Massachusetts General Hospital, Boston, Massachusetts; ²Harvard Medical School, Boston, Massachusetts; ³Division of Gastroenterology, Brigham and Women's Hospital, Boston, Massachusetts; ⁴Department of Nutritional Sciences, King's College London, London, United Kingdom; ⁵Gastroenterology Department, Centre de Recherche Saint-Antoine, Saint-Antoine Hospital, Assistance Publique-Hôpitaux de Paris, Institut National de la Santé et de la Recherche Médicale, Sorbonne Université, Paris, France; ⁶Paris Center for Microbiome Medicine, Fédération Hospitalo-Universitaire, Paris, France; and ⁷Micalis Institute, AgroParisTech, Institut National de Recherche pour l'Agriculture, l'Alimentation et l'Environnement, Université Paris-Saclay, Jouy-en-Josas, France

Inflammatory bowel disease (IBD) comprises chronic and relapsing disorders of the gastrointestinal tract, characterized by dysregulated immune responses to the gut microbiome. The gut microbiome and diet are key environmental factors that influence the onset and progression of IBD and can be leveraged for treatment. In this review, we summarize the current evidence on the role of the gut microbiome and diet in IBD pathogenesis, and the potential of microbiome-directed therapies and dietary interventions to improve IBD outcomes. We discuss available data and the advantages and drawbacks of the different approaches to manipulate the gut microbiome, such as fecal microbiota transplantation, next-generation and conventional probiotics, and postbiotics. We also review the use of diet as a therapeutic tool in IBD, including the effects in induction and maintenance, special diets, and exclusive enteral nutrition. Finally, we highlight the challenges and opportunities for the translation of diet and microbiome interventions into clinical practice, such as the need for personalization, manufacturing and regulatory hurdles, and the specificity to take into account for clinical trial design.

Keywords: Crohn's; Diet; Nutrition; Microbiome; Probiotic.

Inflammatory bowel diseases (IBDs), comprising Crohn's disease (CD) and ulcerative colitis (UC), affect millions of individuals globally. They are chronic, progressive diseases that arise due to a dysregulated immune response to a dysbiotic gut microbiome in genetically susceptible individuals.¹ While the initial phase of investigation of the biology of these diseases focused on genetic variants, we have learned that both the external environment including diet² as well as the internal microenvironment in the form of the gut microbiome³ play a critical role in these disease. While still in early phases, interventional studies modulating diet or targeting the microbiome have shown promise and are likely to be an important part of our armamentarium in managing IBD. In this review, we examine the evidence behind the role of diet and the microbiome in the development and progression of IBD. We define some of the gaps in the existing literature and suggest

solutions toward achieving a goal of personalized diet- and microbiome-based approaches to treat or prevent IBD.

The Microbiome

Microbiome Alterations and Role in IBD Pathogenesis

The gut microbiome represents the complex ecosystem of microorganisms colonizing the human gut. It plays a major role in human physiology and is under the influence of both host and environmental factors, including diet. Until recently, it was not possible to resolve whether the gut microbiome and its alterations were a true actor in the inflammatory process or an inactive witness of the disease. Although some alterations in the gut microbiota observed in IBD are likely to be secondary events related to intestinal inflammation, there are now sufficient data to affirm that microbiota alterations, whether as primary or secondary events, play an active role in the pathogenesis of the disease. Numerous IBD susceptibility genes are involved in innate and adaptive immune responses to microbes,⁴ the luminal content is necessary to induce colitis in mice models and postoperative recurrence in CD,⁵ and antibiotics have some efficacy in specific clinical settings.⁶ However, some decisive arguments have emerged in recent years, including the fact that (1) alterations in the

Abbreviations used in this paper: AID, anti-inflammatory diet; CD, Crohn's disease; CDED, Crohn's disease exclusion diet; CDI, *Clostridioides difficile* infection; CI, confidence interval; EDIP, empiric inflammatory dietary potential; EEN, exclusive enteral nutrition; FMT, fecal microbiota transplantation; FODMAP, Fermentable oligosaccharides, disaccharides, monosaccharides and polyols; FOS, fructo-oligosaccharides; HR, hazard ratio; IBD, inflammatory bowel disease; MDP, Mediterranean diet pattern; PUFA, polyunsaturated fatty acid; RCT, randomized controlled trial; SCD, specific carbohydrate diet; SCFA, short-chain fatty acid; UC, ulcerative colitis; UPF, ultraprocessed food.



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and deconjugation of primary conjugated bile acids produced in the liver by gut microbes into secondary unconjugated is altered in IBD.¹⁸ Finally, the transformation of tryptophan, an essential amino acid, into indole and indole derivatives is a crucial function of the gut microbiome that is altered in IBD. Indeed, several of these tryptophan metabolites are agonists of the aryl hydrocarbon receptor and promote intestinal homeostasis, healing, and immunoregulatory phenotype.¹⁹

Microbiome Interventions: Prebiotics and Probiotics

Given the central role of the microbiome in IBD, altering the microbiome to modify IBD outcomes is a therapeutic area of interest.

Prebiotics are substrate that are selectively utilized by host microorganisms conferring a health benefit. The most common include fructo-oligosaccharides (FOS), galacto-oligosaccharides, and inulin. There have been several notable findings in animal models of IBD.^{20,21} Oligosaccharides increase production of some of the SCFAs, either directly or through cross-feeding, which are known to exhibit anti-inflammatory properties in the gut.²² Daily administration of FOS and inulin were shown to decrease mucosal proinflammatory cytokines in dextran sulfate sodium models. Human data, however, have been mixed. Several small human trials have had notable findings. Lindsay et al²³ assessed 10 patients with ileocolonic CD that were dosed with 15 g of FOS for a 3-week period. In this study, a decrease in Harvey-Bradshaw index and an increase in fecal bifidobacteria was appreciated. Another trial in both CD and UC randomized participants to microencapsulated sodium butyrate or placebo for 2 months.²⁴ In both CD and UC patients, treatment was associated with an increase in SCFA-producing bacteria. However, a study in 103 patients showed that inulin/oligo-fructose did not impact disease activity in CD,²⁵ and other smaller studies have not shown conclusive evidence of benefit in UC.²⁶ There has been a double-blind, placebo-controlled trial of FOS in patients with active CD.²⁵ The primary outcome assessed was clinical response at week 4; the study found no difference in clinical response between the FOS and placebo groups (22% vs 39%) and no differences in fecal *Bifidobacteria* or *F prausnitzii* abundance.

When administered in adequate amounts, probiotics, which are defined as live microorganisms, confer a health benefit to the host. They are generally intended for a healthy population and are regulated as dietary supplements. Probiotics have been assessed as possible therapeutic agents in IBD, particularly in UC. Many trials have focused on induction of remission of UC with various probiotics and results have been mixed. Positive associations have been seen with bifidobacterium. A 2017 meta-analysis identified no significant effect of probiotics overall, but studies of the probiotic VSL#3 showed efficacy in inducing remission compared with placebo (rate ratio, 0.74; 95% confidence interval [CI],

0.63–0.87).²⁷ A 2021 meta-analysis identified 20 randomized controlled trials (RCTs) of probiotics and 5 RCTs of synbiotics for UC.²⁸ In this meta-analysis, a statistically significant reduction in UC disease activity index was found for the effects of probiotics, prebiotics, and synbiotics combined (standardized mean difference, 1.00; 95% CI, 0.27–1.73) with significant heterogeneity between studies. There was no significant effect of probiotics alone on UC activity, though 3 RCTs of 2- to 3-strain probiotic blends (*Lactobacillus*, *Bifidobacterium* strains) did show benefit for reducing UC activity. Other strains such as *E. coli* Nissle 1917 and *Lactobacillus* did not show improved outcomes when compared with mesalamine.²⁹ Tolerance, however, may also be an issue with probiotics in the population, as they can lead to increased gas and bloating as well.

Fewer studies have assessed the efficacy of probiotics for the induction of remission in CD, and most are very small and underpowered. Schultz et al³⁰ performed a double-blind RCT to investigate oral *Lactobacillus* GG compared with placebo for the induction of remission in CD. In this study, only 5 of 11 patients completed the study, with no significant differences appreciated between groups.

Pouchitis is also an area within IBD in which the utility of probiotics has been explored. According to the American Gastroenterological Association Pouchitis guidelines, there is no recommendation for the use of probiotics for the primary prevention of pouchitis given the lack of data. In patients with recurrent episodes of pouchitis that is responsive to antibiotics, the guidelines recommend the use of probiotics to prevent recurrent pouchitis.³¹ Specifically, there were 3 RCTs assessing the efficacy of De Simone formulation of multistrain probiotics for the prevention of recurrent pouchitis; a meta-analysis of these trials found an 87% lower risk of relapse over 12 months.³¹ There is no recommendation regarding the use of probiotics to treat active pouchitis.

Postbiotics are composed of microbial components and metabolites to confer a health benefit. These include metabolites such as SCFAs and secondary bile acids. SCFAs are known to have anti-inflammatory benefits and improve barrier function. Butyrate, delivered as an enema preparation, has been trialed in UC, but most studies failed to show a treatment difference between SCFAs compared with placebo.³² In this trial, 38 patients with distal UC were randomized to receive nightly butyrate enemas or saline. In this study, clinical improvement was noted in 37% of the butyrate-treated patients and in 47% of the placebo-treated patients. Secondary bile acids, which are the byproduct of microbial metabolism of primary bile acids, reduce inflammation in animal models; however, there is limited use in human IBD cohorts.³³

Microbiome Interventions: FMT

FMT has become a widely utilized method to restore the entire microbial ecosystem and not just gut bacteria.

While traditionally this therapy was reserved for multiply recurrent *Clostridioides difficile* infection (CDI), more recently it has been explored as a preventative strategy earlier in the treatment paradigm for this complication.³⁴ Additionally, there were initial concerns regarding potential IBD worsening when used to prevent recurrent CDI in patients with IBD. However, in a prospective study it was found to be safe and yields high cure rates,³⁵ with low rates of disease exacerbation following this initial concern.³⁶

Studies of FMT in IBD for non-CDI indications have focused primarily on UC (Table 1). A systematic review of 10 RCTs concluded that FMT is effective for inducing clinical and endoscopic remission of mild-to-moderately active UC without a significant increased risk of adverse events.³⁶ There is a paucity of data on the use of FMT for maintenance of remission of UC after FMT. However, 1 trial noted maintenance of remission through week 56 among participants who achieved remission with FMT capsules after antibiotic pretreatment; this was not observed in patients not receiving further FMT after induction.⁴⁴ Another trial randomized 73 patients with mild-to-moderate UC to standard medical therapy or weekly FMT (derived from rural donors and delivered via a colonoscopy) for 7 weeks, combined with an anti-inflammatory diet.⁴⁵ Participants receiving FMT arm had significant high rates of both clinical and endoscopic remission at weeks 8 and 48.

The data in CD are much more limited, and this likely reflects the heterogeneous nature of CD and difficulty conducting trials in this population. A single RCT of 17 patients with active CD receiving corticosteroids randomized them to a colonoscopic FMT or a sham FMT.⁴⁶ At week 10, steroid-free clinical remission rates were numerically higher in the FMT group. Two trials of FMT for pouchitis using either a sham FMT⁴⁷ or autologous FMT⁴⁸ showed no benefit. A small pilot study in 10 patients with IBD with concurrent primary sclerosing cholangitis showed improvement of alkaline phosphatase in 3 of 10 patients.⁴⁹

More recently, the Food and Drug Administration has introduced a new term, live biotherapeutic products. These are defined as biological products that contain live organisms applicable to the prevention, treatment, or cure of a disease, and are also being studied to treat UC. These differ from traditional probiotics in that there is a pharmaceutical expectation and they are intended to be used in a sick population. These are regulated as drugs and therefore have been through phase 2 and 3 studies.⁵⁰ Additionally, there is ongoing debate as to whether products that are donor derived should be termed “live biotherapeutic products” as the Food and Drug Administration has referred to donor-derived products as “fecal microbiota products” more recently. Two live biotherapeutic products, live-jslm (Rebyota) and fecal microbiota spores, live-brpk (VOWST), are now approved for the prevention of recurrent CDI,^{51,52} with indications beyond CDI being the next frontier. In UC, the

first trials of an investigational fecal microbiota product, SER-287, showed efficacy in a phase 1 trial but failed to meet endpoints in a phase 2b study.^{53,54} VE202 is a synthetic (not donor-derived) bacterial consortium containing a mixture of spore-forming *Clostridia* strains⁵⁵ and is currently being evaluated in a phase 2 program for the treatment of biologic-naïve mild-to-moderate UC (NCT05370885). EXL01 is a single *F. prausnitzii* strain that is currently evaluated in the maintenance of steroid-induced clinical response/remission in patients with ileal CD (NCT05542355). Other fecal microbiota products are in early-phase development for UC as well. This therapeutic area is rapidly evolving and holds much promise.

Diet

Diet and Risk of Incident IBD

Exposures early in life, the period of maximum vulnerability of the microbiome and developing immune system, modifies risk of IBD. In a systematic review of population-based studies, a history of being breastfed in infancy was associated with a lower risk of CD (odds ratio, 0.71; 95% CI, 0.59–0.85) and UC (odds ratio, 0.78; 95% CI, 0.67–0.91).⁵⁶ In an elegant study, Guo *et al*⁵⁷ examined diet at 1 year to examine association with future onset IBD in Swedish and Norwegian birth cohorts. Compared with infants with a low diet quality at 1 year (assessed using the Healthy Eating Index), children with high diet quality had a reduced risk of IBD (hazard ratio [HR], 0.75; 95% CI, 0.56–1.00). Specific food constituents associated with this included an inverse association with fish and vegetable intake, while intake of sugar-sweetened beverages was associated with a higher risk of IBD.

Much of the initial attempts at identifying dietary risk factors focused on macronutrients. In a prospective cohort of 170,805 women, a higher intake of long-chain n-3 polyunsaturated fatty acids (PUFAs) comprising docosapentaenoic acid, eicosapentaenoic acid, docosahexaenoic acid (0.72; 95% CI, 0.51–1.01), or a lower n-3/n-6 PUFA ratio was associated with reduced risk of UC.⁵⁸ Conversely, high intake of arachidonic acid, an n-6 PUFA, and the concentration of arachidonic acid in gluteal fat pad biopsies was associated with increased risk of UC (rate ratio, 4.16; 95% CI, 1.56–11.04).⁵⁹ Studies have also suggested that this association may be modified by host genotype, in particular polymorphisms at the *CYP4F3* and *FADS2* loci.⁶⁰ Initial evidence from both case-control studies and prospective cohorts suggested an inverse association between dietary fiber and risk of IBD. In an analysis of the Nurses' Health Study cohorts, women in the highest quintile of intake of dietary fiber (median of 24 g/d) had a lower risk of development of CD than those in the lowest quintile (HR, 0.59; 95% CI, 0.39–0.90). Adding nuance to the impact of fiber on risk of IBD is the potential differential effect by subtype of fiber. For example, in the Nurses' Health Study cohorts,

Table 1. Summary of Randomized Controlled Trials of FMT for the Treatment of Ulcerative Colitis

Study	n	Disease Activity for Inclusion	FMT Route	Number of Infusions	Dosage	Stool Donor	Comparator	Primary Endpoints	Follow-Up	Clinical Remission	Clinical Response
Moayyedi et al, 2015 ³⁷	65	Mayo clinical score ≥ 4 and endoscopic score ≥ 1	Enema (50 mL)	6 (weekly)	50 g stool in 300 mL water	Unrelated donor	Water enema	Total Mayo score ≤ 2 and endoscopic Mayo = 0	7 wk	FMT group: 9/38, control group: 2/37, $P = .03$	FMT group: 15/38, control group: 9/37, $P = .16$ (≥ 3 decrease in total Mayo score)
Rossen et al, 2015 ³⁸	48	SCCAI ≥ 4 and ≤ 11	Nasoduodenal (500 mL)	2 (weeks 0 and 3)	Minimum 60 g stool in 500 mL saline	Patient directed donor	Autologous stool	SCCAI ≤ 2 - and ≥ 1 -point decrease in Mayo endoscopic score	12 wk	FMT group: 7/23, control group: 5/25, $P = .29$	FMT group: 11/23, control group: 13/25, $P = \text{NS}$
Paramsothy et al, 2017 ³⁹	81	Mayo clinical score ≥ 4 and ≤ 10 and endoscopic subscore ≥ 1	Initial colonoscopy (150 mL) then enemas	40 (5 per week for 8 wk)	37.5 g stool in 150 mL	Mixed unrelated donors (3–7 donors per patient)	Saline enema	Steroid free total Mayo score ≤ 2 , with all subscores ≤ 1 , and ≥ 1 -point decrease in Mayo endoscopic score	8 wk	FMT group: 11/41, control group: 3/40, $P = .02$	FMT group: 22/41, control group: 9/40, $P = .01$
Costello et al, 2019 ⁴⁰	73	Mayo clinical score ≥ 3 and ≤ 10 and endoscopic subscore ≥ 2	Initial colonoscopy (150 mL) then enemas	3 (week 0 colonoscopy, 2 enemas week 1)	Colonoscopy 50 g stool in 200 mL saline and glycerol, enema 25 g stool in 100 mL	Mixed unrelated donors (3–4 donors per patient)	Autologous stool	Steroid-free total Mayo score ≤ 2 and Mayo endoscopic score ≤ 1	8 wk	FMT group: 12/38, control group: 3/35, $P = .03$	FMT group: 21/38, control group: 8/35, $P < .01$ (≥ 3 decrease in total Mayo score)
Brezina et al, 2021 ⁴¹	45	Mayo clinical score ≥ 4 and ≤ 10 , left sided disease only	Enema (150–170 mL)	10 (5 in first week and once weekly for 5 wk)	50 g stool in 150 mL saline and glycerol	Unrelated donor	5-ASA enema	Total Mayo score ≤ 2 with no subscore > 1 at week 12	1 y	FMT group: 12/21, control group: 8/22, 95% CI: –7.6%, 48.9%	FMT group: 15/21, control group: 12/22, $P = .35$
Crothers et al, 2021 ⁴²	12	Mayo clinical score ≥ 4 and ≤ 10 and all subscores ≥ 1	Initial colonoscopy (120 mL) then daily colonoscopic FMT	1 (induction) + daily for 12 wk	1 g stool/2.5 mL in colonoscopy infusion, 0.5 g stool per FMT capsule	Mixed unrelated donors (2 donors for entire study)	Sham therapy	Safety and feasibility of long-term colonoscopic FMT maintenance therapy	36 wk	Colonoscopic FMT group: 2/6, placebo group: 0/6; 95% CI, = 0.38-infinity, $P = .45$	Colonoscopic FMT group: 3/6, placebo group: 1/6; 95% CI, = 0.42–21.20, $P = .55$
Fang et al, 2021 ⁴³	20	Mayo clinical score ≥ 4 and ≤ 12	Colonoscopy (150 mL)	Once	50 g stool in 150 mL saline	Unrelated donor	Mesalamine or corticosteroid induction with mesalamine maintenance	Mayo score ≤ 2 with each subscore ≤ 1 , Mayo endoscopic subscore ≤ 1 compared with baseline	24 mo	FMT group: 9/10, control group: 5/10, $P = .019$	FMT group: 10/10, control group: 10/10, $p = 0.000$ (compared with pretreatment)

Table 1. Continued

Study	n	Disease Activity for Inclusion	FMT Route	Number of Infusions	Dosage	Stool Donor	Comparator	Primary Endpoints	Follow-Up	Clinical Remission	Clinical Response
Hafler et al, 2022 ⁴⁴	35	Total Mayo score ≥ 4 and ≤ 10 and endoscopic subscore ≥ 1	Oral capsule	24 capsules per day for 1 wk, 12 per day for 1 wk, 6 per day for 6 wk	0.35 g stool per FMT capsule	Unrelated donor	Placebo capsule	Steroid-free total Mayo score ≤ 2 , all subscores ≤ 1 , and ≥ 1 -point reduction in endoscopic subscore	8 wk	FMT group: 11/15, placebo group: 5/20, $P = .005$	FMT group: 11/15, placebo group: 9/20, $P = .94$
Kedia et al, 2022 ⁴⁵	66	SCCAI ≥ 3 and ≤ 9 , UCEIS ≥ 1	Colonoscopy (200–250 mL)	Once	50 g stool in 200–250 mL saline	Mixed unrelated donors (2–5 per patient)	Standard medical therapy	SCCAI < 2 and UCEIS < 1	48 wk	FMT group: 21/35, SMT group: 10/31, $P = .02$	FMT group: 23/35, SMT group: 11/31, $P = .01$

Values are n/n, unless otherwise indicated. SCCAI, simple clinical colitis activity index; SMT, standard medical therapy; UCEIS, Ulcerative Colitis Endoscopic Index of Severity. CI, confidence interval; FMT, fecal microbiota transplantation.

an inverse association with CD was stronger for fiber intake from fruits, while fiber intake from whole grains, legumes, or cereal was not associated with modified risk.⁶¹ Different sources and subtypes of fiber may differ by solubility of fiber, fermentation potential, and content of other potentially bioactive constituents such as polyphenols. There are less consistent data on the association between other macronutrients and risk of IBD. An analysis of the EPIC (European Prospective Investigation into Cancer and Nutrition) cohort showed an increase in risk of IBD with higher intake of animal (but not plant) protein.⁶²

Studies have also examined the association between dietary micronutrients and risk of IBD. An increase in intake of dietary zinc was associated with reduced risk of IBD within the Nurses' Health Study cohorts,⁶³ an association mechanistically supported by a beneficial impact of zinc on epithelial barrier in experimental models. While in experimental studies, oral iron was associated with a proinflammatory effect, and this association has been less robustly found in human cohorts with variable to null association between dietary iron or heme intake and risk of IBD.⁶⁴ Antioxidants in diet may play a role in modifying inflammation. In a study by Lu et al,⁶⁵ flavones and resveratrol in the diet were associated with a reduced risk of CD. Food additives may also play a role in the pathogenesis of IBD. In the Nurses' Health Study cohorts, compared with participants with low ultra-processed food (UPF) consumption, those in the highest quartile of intake had an increased risk of CD (HR, 1.70; 95% CI, 1.23–2.35)⁶⁶; however, an association with risk of UC was not found. Specifically, ultraprocessed breads and breakfast foods, sauces, cheeses, spreads, and gravies showed the strongest association with CD risk.

Food is consumed within the context of other dietary items, and consequently, dietary patterns may be more relevant in informing dietary strategies for the prevention and treatment of IBD, rather than individual food group association. In a prospective cohort of 83,147 Swedish men and women, adherence to a Mediterranean diet pattern (MDP) was associated with a lower risk of CD (HR, 0.42; 95% CI, 0.22–0.80) but not UC.⁶⁷ An empiric inflammatory dietary potential (EDIP) derived from the association of foods with inflammatory cytokines in the serum was associated with a 51% higher risk of CD (HR, 1.51; 95% CI, 1.10–2.08) in 2 prospective cohorts.⁶⁸ Interestingly, compared with participants who were in the lowest tertile of EDIP score at baseline who subsequently moved to the highest tertile 8 years later, there was a 2-fold increase in relative risk of CD (HR, 2.05; 95% CI, 1.10–3.79), although the inverse diet, a reduction in risk on change from a high to low EDIP score, was not observed.

Diet and Disease Course

There have been fewer studies examining dietary risk factors for disease relapse or other outcomes. Two

studies suggested that increased red meat intake was associated with higher risk of disease flares in patients with UC, while 2 others did not find such an association.^{69,70} In the PREDiCt (Prognostic Effect of Environmental Factors in Crohn's and Colitis) study that followed 520 patients with UC (and 497 patients with CD), patients with UC in the highest reported meat intake had a 2-fold increase in risk of flares compared with those in the lowest category.⁷⁰ In contrast, in an Internet-based trial of red meat intake in the IBD Partners cohort, consumption of 2 or more servings of red meat per week was not associated with an increased risk of disease flares.⁶⁹ Similarly, there are inconsistent data on whether dietary fiber may play a role in preventing relapse. Two studies showed an inverse association between fiber intake and risk of IBD relapse, while 2 studies found no benefit and 1 study demonstrated an increase in risk of disease flares with high fiber intake.^{71,72} There are several potential explanations for these divergent findings. First, studies of association of fiber intake and relapse are observational, and regular flares may alter dietary intake, rather than the other way around. This could also be explained by differences in source and subtypes of fiber as highlighted previously. In addition, an individuals' microbiome may influence their ability to process fiber and generate biologically active metabolites, leading to biology-dependent heterogeneity in effect.⁷³ Finally, studies have also highlighted the generation of beneficial SCFAs from fermentable fiber may be based on inflammatory status, leading to differing effects on the microbiome and metabolome based on presence of active disease.⁷⁴ Diet may influence outcomes beyond disease relapse alone. Godny et al⁷⁵ found that a low intake of fruits was associated with increased occurrence of pouchitis. Mechanistically, fruit consumption was associated with a favorable microbial profile, and influences abundance of *Faecalibacterium*, *Lachnospira*, and *Ruminococcus*.

Diet in the Treatment and Maintenance of IBD. Patients with IBD are often very keen to attempt dietary approaches for the induction and maintenance of remission, and doctors and dietitians should use evidence-based practice when advising and supporting patients through dietary management (Figure 2).

Exclusive Enteral Nutrition. The most widely investigated dietary intervention in IBD is exclusive enteral nutrition (EEN). This involves the oral or enteral delivery of nutritionally complete formula as a sole source of nutrition for 4 to 8 weeks with the exclusion of all foods and drinks, with some regimens allowing some clear fluids. EEN has largely been examined as treatment for active disease with evidence of benefit for CD but not for UC. A Cochrane review of 10 trials comparing EEN with corticosteroids showed greater benefit in pediatric than adult CD, and in per-protocol over intention-to-treat analysis (Table 2).⁷⁶ In 2 pediatric trials, there was no difference in remission rates between EEN and corticosteroids on intention-to-treat analysis; however, on per-

protocol analysis (excluding children who could not complete treatment due to poor palatability or lack of acceptance of a nasogastric tube), EEN (89%) resulted in superior remission rates compared with corticosteroids (61%). In view of these findings, EEN is often first-line treatment in pediatric CD, as it also spares children from early-life corticosteroid exposure. Attrition from EEN therapy is high in both research and clinical practice⁷⁶ and is frequently due to issues of palatability, inability to consume the volumes required, and/or tolerance of a nasogastric tube. Identifying responders and nonresponders to EEN could better target who should be offered this therapy. A recent study showed EEN responders had higher gut microbiota richness and lower stool acetate and butyrate, with a multicomponent model incorporating microbiome, metabolome, diet, clinical, and immunological data predicting response to EEN with 78% accuracy.⁷⁷ In view of poor tolerance to EEN, attempts have been made to circumvent its use through dietary management approaches that partially (eg, CD exclusion diet [CDED]) or totally (CD-TREAT) replace EEN with normal foods. Despite the established efficacy of EEN or EEN-based diets, certain paradoxes remain unexplained with this treatment option. The microbial changes induced by EEN such as reduced diversity contradict the prevailing view of a healthy microbiome. In addition, EEN formulations have a high content of additives, which contrasts with therapeutic diets that attempt to minimize additive- and emulsifier-containing foods.⁷⁸ Further research is required to examine whether this is influenced by variation in biologic effect by type of additive.

CD Exclusion Diet. The CDED restricts a range of foods including dairy products gluten, baked goods and breads, processed meats, animal fats, canned/packaged products, emulsifiers, and other food additives.⁷⁹ The mechanistic rationale was based on in vitro and animal evidence of a possible impact on the gut microbiome and gut permeability.

Numerous uncontrolled clinical evaluations have been performed, but the most significant study was a randomized comparative trial of CDED as an adjunct to partial enteral nutrition compared with EEN in 74 children with active CD.⁸⁰ After 6 weeks, the primary outcome of tolerance was higher following CDED and partial enteral nutrition (97.5%) than EEN (73.7%) ($P = .002$), but there were no differences in clinical outcomes including clinical response and remission. At 12 weeks, during which phased reintroduction of normal foods occurred in the EEN group, the CDED plus partial enteral nutrition group had higher rates of sustained remission than the former EEN group.⁸⁰ In another randomized comparative trial, CDED plus partial enteral nutrition was compared with CDED alone in 40 adults with active CD.⁸¹ There were no differences in the primary outcome of remission at 6 weeks between CDED plus partial enteral nutrition (68.4%) and CDED (57.1%) ($P = .4618$), nor were there any differences in response

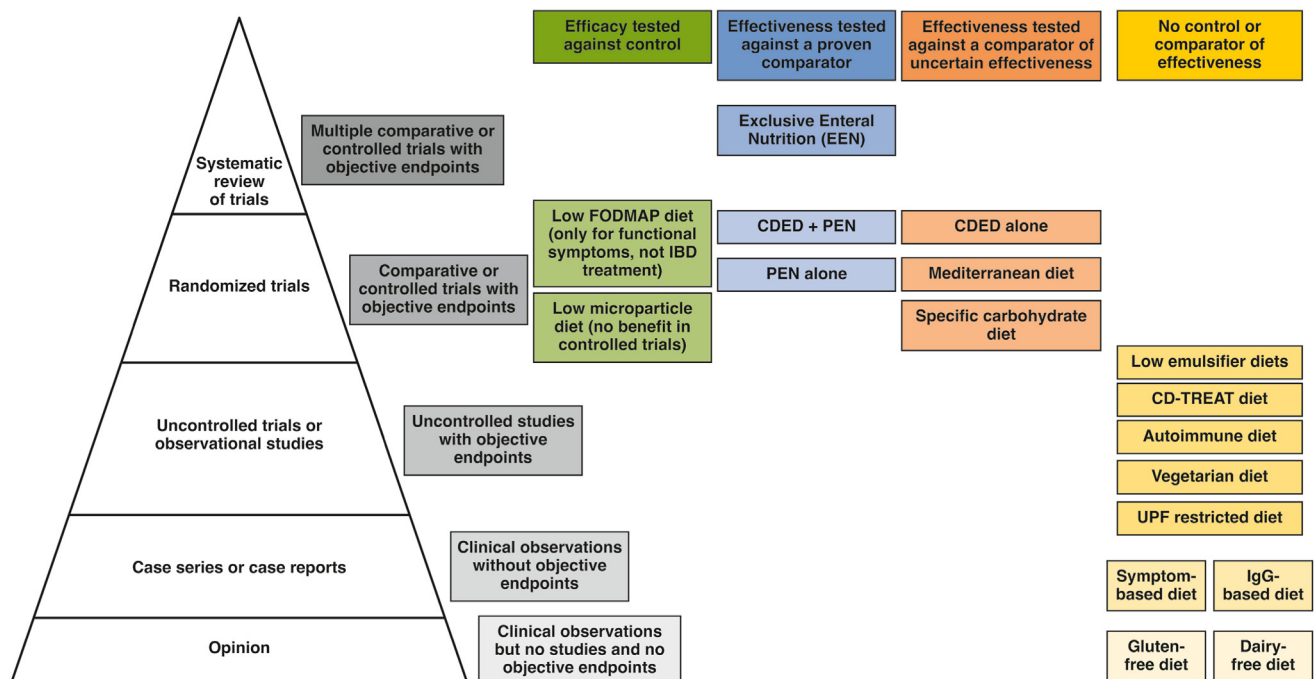


Figure 2. Evidence ladder for studies that have tested the efficacy or effectiveness of diet in the management of IBD. The hierarchy defines the levels of evidence used in evidence-based practice and is adapted here specifically for trials of diet in IBD including the use of objective inflammatory endpoints. Few randomized controlled trials have been conducted to demonstrate efficacy due to the challenges of control diet interventions (green). More randomized comparative trials have been conducted that compare diet with another intervention that is proven to be effective (blue) or with an intervention with uncertain effectiveness (orange). Many diets have not been tested in clinical trials with control or comparator groups and are based on uncontrolled trials, observational studies, case series, or case reports or opinion (yellow). PEN, partial enteral nutrition.

or remission at any of the time points (6, 12, or 24 weeks).

Other Therapeutic Diets. Several other diets have been examined as therapeutic options for the management of CD. The CD-TREAT diet was developed to replace EEN by mimicking the effect of enteral formulas with everyday foods, and therefore the diet excludes gluten, lactose, and alcohol and restricts starches and fiber to better match the profile of commercial formulas. The CD-TREAT diet was tested in 25 healthy adults and

in HLA-B7 B27 transgenic rats, showing moderate-to-strong correlation for changes in both operational taxonomic units and genera between CD-TREAT and EEN.⁸² In a pilot study in 5 children with active CD, 4 completed an 8-week treatment with CD-TREAT, all of whom entered remission, with statistically significant reductions in disease activity scores and fecal calprotectin.⁸² In the Diet to INduce Remission in Crohn's disease (DINE-CD) study of mildly symptomatic patients with CD, an MDP (high in fruits, vegetables, legumes,

Table 2. Summary of Findings From the Cochrane Meta-Analysis of Clinical Trials of EEN vs Corticosteroids for the Induction of Remission During the Treatment of Active Crohn's Disease

	Analytical Approach ^a	Trials/Patients	Remission Rate		Risk of Remission in EEN	P Value
			EEN	Corticosteroids		
Children	Intention to treat	2/57	83 (24/29)	61 (17/28)	1.35 (0.92–1.97)	.13
	Per protocol	2/55	89 (24/27)	61 (17/28)	1.43 (1.03–1.97)	.031
Adults	Intention to treat	6/352	45 (87/194)	73 (116/158)	0.65 (0.52–0.82)	.0002
	Per protocol	6/307	58 (87/149)	73 (116/158)	0.82 (0.70–0.95)	.0092
All patients	Intention to treat	8/409	50 (111/223)	72 (133/186)	0.77 (0.58–1.03)	.08
	Per protocol	8/362	63 (111/176)	72 (133/186)	0.93 (0.75–1.14)	.47

Values are n, %, (n/n), or risk ratio (95% confidence interval). Data were extracted from Narula et al.⁷⁶

^aIntention-to-treat analysis included all patients who started the trial. Per-protocol analysis excluded treatment failures due to poor palatability or lack of acceptance of a nasogastric tube (other treatment failures still included).

nuts and fish) resulted in symptomatic improvement in nearly half the individuals and resulted in improved C-reactive protein or fecal calprotectin in up to a third of patients with elevated levels at baseline. In a prospective trial of 28 patients with quiescent UC treated with the MDP or control Canadian habitual diet for 12 weeks, a smaller fraction of patients in the MDP arm (20%) had elevated fecal calprotectin at the end of the study compared with the Canadian habitual diet (75%).⁸³ In addition, the MDP induced several favorable alterations in the gut microbiome. A 6-month open-labeled intervention with an anti-inflammatory diet (AID) designed to increase dietary fiber, antioxidants, and n-3 PUFAs while reducing red meat, processed meat, and sugar in patients with UC in clinical remission resulted in a lower fraction of relapse and larger fraction with normal fecal calprotectin at the end of the intervention in the AID group.⁸⁴ Other variations of the anti-inflammatory diet⁸⁵ and autoimmune protocol diets⁸⁶ have also demonstrated symptom improvement in case series but lack rigorous clinical trial evidence in support of efficacy. A dietary strategy to reduce hydrogen sulfide production (4-SURE [4 Strategies to Sulfide REduction]) diet was tolerable and acceptable in 28 adults with UC, with clinical response in nearly half the patients (46%) and endoscopic improvement in one-third (36%).⁸⁷ The diet was accompanied by an elevation in fecal SCFA content.

Specific Carbohydrate Diet. The specific carbohydrate diet (SCD) was developed 100 years ago and aims to restrict disaccharides and polysaccharides that are incompletely or completely unabsorbed in the gastrointestinal tract and are available for microbial fermentation. The diet excludes grains, sugars, and milk products, although it is incompletely defined, with inconsistencies such as fruits and vegetables being permitted despite being rich sources of polysaccharides. The mechanistic rationale for the SCD is not fully evidenced, and until recently, clinical evidence for the SCD was limited to case series, until the publication of 2 clinical trials. The first was a randomized comparative trial comparing SCD, a modified SCD protocol, and a whole food diet for 12 weeks.⁸⁸ Of the 18 children with active CD who enrolled, only 10 completed the study, all achieving clinical remission.⁸⁸ More recently, the previously described DINE-CD study compared an SCD with an MDP in 194 adults with mild/moderate symptoms of CD.⁸⁹ At 6 weeks, there was no difference in the primary outcome of symptomatic remission (SCD 46.5% vs MDP 43.5%), fecal calprotectin, or CRP response. The lack of control group limited the ability efficacy to infer effectiveness of either diet. Most clinical studies of SCD did not mandate significant objective inflammation at inclusion, and while biomarkers were measured at follow-up, the primary clinical outcome was not contingent on demonstrating resolution objectively of inflammation. Thus, while there is evidence of symptom benefit with SCD, to what degree this is accompanied by resolution of objective inflammation remains to be robustly established.

UPFs and Food Additives. Most therapeutic diets in IBD, CDED, CD-TREAT, MDP, or SCD, advise on the restriction of processed foods as part of other dietary restrictions; thus, the effect of UPF restriction alone cannot be ascertained. In addition, such trials did not quantitatively measure UPF intake. Food additives are substances intentionally added to food for a technological purpose and some, including emulsifiers, sweeteners, and colorants, have been shown to modify the microbiome and impact epithelial integrity in animal models.⁹⁰ Research in people with IBD has only been conducted for emulsifiers, albeit in small clinical trials. A crossover trial in 7 patients with quiescent UC demonstrated carrageenan supplementation did not induce gut symptoms,⁹¹ while a resupplementation trial in 12 patients with UC showed carrageenan-restricted diet reduced relapse rates,⁹² and an uncontrolled trial in CD showed restriction of all emulsifiers improved gut symptoms in 20 people.⁹³

Low FODMAP Diet. Fermentable oligosaccharides, disaccharides, monosaccharides and polyols (FODMAPs) are incompletely digested carbohydrates, some of which increase small intestinal water or colonic gas production. Temporary dietary restriction of FODMAP, followed by staged reintroduction to tolerance, is termed the low FODMAP diet and has been extensively studied in irritable bowel syndrome. An RCT in 52 people with IBD demonstrated the low FODMAP diet reduced bloating, flatulence, and stool frequency compared with a sham diet.⁹⁴ Importantly, the low FODMAP diet should not be used to treat active inflammation in IBD, but rather to manage burdensome gut symptoms in the absence of active disease.

Translation of Diet and Microbiome Interventions: Challenges and Opportunities

There are many common challenges to translating diet- and microbiome-based interventions in IBD and opportunities to stream-line interventional studies.

1. **Defining mechanism of action:** Unlike small molecules and biologics in which the precise immunologic target is clearly known, the biologic mechanisms of influence of both diet- and microbiome-based interventions are less well defined. Unlike measurable drug levels, at this time, there are also no reliable measures to quantify pharmacokinetic effect of diet- or microbiome-based interventions. This impedes identification of an effective dose, optimizing treatment, or monitoring biologic effect. There is an important need in the field for the development of quantitative biomarkers that can measure the biologic impact of these interventions to help precisely develop such interventions.
2. **Biologic heterogeneity and personalization:** There is wide heterogeneity in an individual's

microbiome composition, impacted by geography and environment, in addition to host factors. Beyond that, there is significant interindividual variability in microbial composition, and within the same individual at different time points. This challenges both the effectiveness and interpretation of microbiome-based intervention studies. Similarly, there is also differential susceptibility to dietary interventions that may be influenced by host genetics (gene-diet interactions) or underlying microbial composition (as has been identified for weight loss interventions). A more robust definition of the parameters contributing to this heterogeneity can help more precisely target the treatments to the subgroup most likely to benefit.

3. **Manufacturing and regulatory hurdles:** The regulatory path for microbiome-based interventions continues to evolve, with significant heterogeneity in regulations between different countries. This has hampered intervention trials across regions internationally. Similar variability in dietary practices across different populations may affect the effectiveness (or acceptability) of dietary interventions. There is also the need for efficient and scalable production of both diet- and microbiome-based interventions and a reproducible and consistent way to deliver such interventions. For microbiome interventions, it may be important to deliver it to the most appropriate region of the gut for benefit. For both dietary and microbiome interventions, it is important to precisely quantify all components of the intervention, particularly for mixed consortia or whole food diet-based treatments.
4. **Optimizing robust clinical trial design:** An important challenge in studies of microbiome- and diet-based interventions is patient selection and study design. Most studies of these interventions have focused on mild-to-moderate CD or UC; thus, it is unclear if such interventions (as opposed to immune-directed therapy) may have a role in those with more severe disease. Diet is a challenging intervention to investigate. It is difficult though not impossible to design “control” interventions, and controlled trials are therefore rare. Comparative trials of diet are often used to inform the best options for clinical practice; however, a trial of 2 diet interventions that does not show differences do not prove efficacy of either and does not prove equivalence unless the study was powered for equivalence analysis. Use of multidisciplinary teams with members with expertise in delivering dietary interventions and maximizing adherence is important to optimize success of such interventional trials. Assessment of adherence also relies primarily on self-report. It is important for research on objective biomarkers that could

measure both adherence to dietary interventions as well as quantify their health impact, accounting for heterogeneity by individual biology. Studies have also frequently used subjective criteria for entry and as an endpoint. To generalizably inform their role in the therapeutic algorithm, it is important to have meaningful objective biomarker- and endoscopy-based endpoints to quantify effectiveness. It is important to examine them as both single and combination (along with conventional treatment) interventions, for varying durations of treatment, and both for induction and for maintenance of remission. Finally, it is likely that trial designs currently used to assess the efficacy of immunosuppressors are not optimal for evaluating microbiome- and diet-based interventions, and there is a need for innovation in this regard.

Conclusions

The external environment, through diet, and the internal gut microenvironment through the microbiome, play a central role in the development and propagation of IBD. While large prospective cohort studies have informed dietary risk factors prior to disease onset, there remains an important need to identify how diet influences established disease contributing to relapse or disease progression. Cross-sectional studies of well-phenotyped cohorts have provided important insights into the microbial architecture of IBD, but further work is necessary to identify longitudinal trajectory of the microbiome in relation to key disease outcomes. Dietary and microbiome-based treatments offer considerable promise in the management and potentially the prevention of IBD. High-quality interventional studies are required in well-characterized populations to inform the positioning of these treatments within our therapeutic algorithm.

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Correspondence

Address correspondence to: Ashwin N Ananthakrishnan, MD, MPH, MGH Crohn's and Colitis Center, 165 Cambridge Street, 9th Floor, Boston, Massachusetts 02114. e-mail: aananthakrishnan@mgh.harvard.edu.

Conflicts of interest

The authors disclose the following: Harry Sokol has received lecture fees from, served on the board for, or served as a consultant for AbbVie, Adare, Amgen, Biocodex, Biogen, Bledina, Bromatech, Celltrion, Ferring, Fresenius, Galapagos, Immusmol, IPSEN, Janssen, Lilly, Mayoli, MSD, Nestlé, Sanofi, Servier, Takeda, Tillotts, Urigo, and Viatrix; owns stock in Enterome Bioscience; and is co-founder of Exelium Biosciences. Jessica R. Allegretti has served as a consultant for Janssen, Pfizer, AbbVie, Finch Therapeutics, Seres Therapeutics, Ferring, GSK, Merck, Bristol Myers Squibb, Roivant, and Adiso; received grant support from Pfizer, Janssen, and Merck; and served as a speaker for BMS, AbbVie, and Janssen. Kevin Whelan has received research grants from Almond Board of California, Danone, and the International Nut and Dried Fruit Council; has received speaker fees from Danone; and is the holder of a joint patent to use volatile organic compounds as biomarkers in irritable bowel syndrome (PCT/GB2020/051604). Ashwin N. Ananthakrishnan has served on the advisory board for Geneoscopy.